Biochemistry Cume – March 8, 2013

1 (50pts). The half-reactions involved in the LDH reaction and their standard reduction potentials are

pyruvate $+2H^+ + 2e^- \rightarrow$ lactate $\quad E^{o'} = 0.185V$

$NAD^+ + 2H^+ + 2e^- \rightarrow NADH + H^+ \quad E^{o'} = -0.315V$

Faraday const= 96,485 J (96.485 kJ) per volt gram equivalent

R = 8.314 J/(mol-K)

A) Calculate ΔG for the reaction at the biochemical standard state under the following conditions:

[lactate]/[pyruvate] = 160. $[NAD^+]$[NADH] = 160.

B) Under what conditions will the reaction spontaneously favor NADH oxidation?

2 [25pts] How many different disaccharides of D-glucopyranose are possible? Justify your answer and explain your reasoning.

3. [25pts] What is the growth rate, in helical turns per second, of the alpha helices in a human hair that is growing 15 cm/year? The amino acids hair are arranged in a right-handed helical structure where each amino acid residue corresponds to a 100° rotation and a translation of 1.5 Å (0.15 nm) along the helical axis.